

$$(2) \text{ Fie } K_m(x, t) = (x-t)_+^m - L_m((x-t)_+^m; x) \\ (x-t)_+^m = \begin{cases} 0, & x \leq t \\ (x-t)^m, & x > t \end{cases}$$

Dacă $f \in C^{(m)}[a, b]$ se are ca:

$$f(x) - L_m(f; x_0, \dots, x_n)(x) = \int_a^b K_m(x, t) f^{(m)}(t) dt$$

(3) Fie $A: D[a, b] \rightarrow \mathbb{R}$ o funcție liniară de formă

$$A(f) = \sum_{k=0}^m c_k f(x_k)$$

unde $D[a, b]$ este mulțimea tuturor funcțiilor definite pe $x_k, x_k \in [a, b], k=0, \dots, n, x_i \neq x_k, i \neq k, c_k \in \mathbb{R}$

Se are ca:

$$A(f) = \sum_{k=0}^m a_k [x_0, x_1, \dots, x_k; f]$$

deci $A(1) = A(x) = 0$. și $a_k = \sum_{j=k}^m c_j (x_j - x_0) \dots (x_j - x_{j-1})$,

(4) Fie $P_n(f)(x) = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} f\left(\frac{k}{n}\right)$. Se are ca:

$$P_n(f)(x) = \sum_{k=0}^n \frac{k!}{n^k} \binom{n}{k} \left[0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{k}{n}; f\right] x^k.$$

(5) Fie $x_i, i=0, \dots, n$ distincte, $0 \leq k \leq n$. Se are ca:

$$[x_0, x_1, \dots, x_n; (x-x_0) \dots (x-x_k) f] = [x_{k+1}, x_{k+2}, \dots, x_n; f]$$

(6) Fie $x_i, i = \overline{0, n}$ puncte distincte, $m \in \mathbb{N}$ $m < n$
 Se se cote ca

$$[x_0, x_1, \dots, x_m, x_{m+1}, \dots, x_n, f] = [x_0, \dots, x_m, \frac{f(x)}{(x-x_{m+1}) \dots (x-x_n)}] + \\ + [x_{m+1}, \dots, x_n, \frac{f}{(x-x_0) \dots (x-x_m)}]$$

(7) Se se calculeze:

$$[x_0, x_1, \dots, x_n, \frac{1}{ax+b}]$$

$$x_i \neq x_j \neq \frac{b}{a}, a \neq 0.$$

(8) Se se demonstreze egalitatea:

$$[x_0, x_1, \dots, x_n, f, g] = \sum_{k=0}^n [x_0, \dots, x_k, f] [x_k, x_{k+1}, \dots, x_n, g]$$

(Formula lui T. Popoviciu)

a) Se se calculeze:

$$a) [0, 1, 2, \dots, n, x^{n+2}] \quad b) [0, 1, 2, \dots, n, \frac{x^{n+2}}{x+1}]$$

(9) Se se cote ca o functie $f: [a, b] \rightarrow \mathbb{R}$ este
 convexa atunci pt orice puncte distincte din $[a, b]$
 $[x_1, x_2, x_3, f] \geq 0$

(10) Se se note volumul de grad minim $\sqrt[n]{\text{con}}$
 indeplinite conditiile: $f^{(k)}(a) = P^{(k)}(a), k = \overline{0, m}, f^{(k)}(b) = Q^{(k)}(b)$
 pt $k = \overline{0, n}$, f fiind o functie derivabila de $\max(m, n)$
 in punctele a, b .